

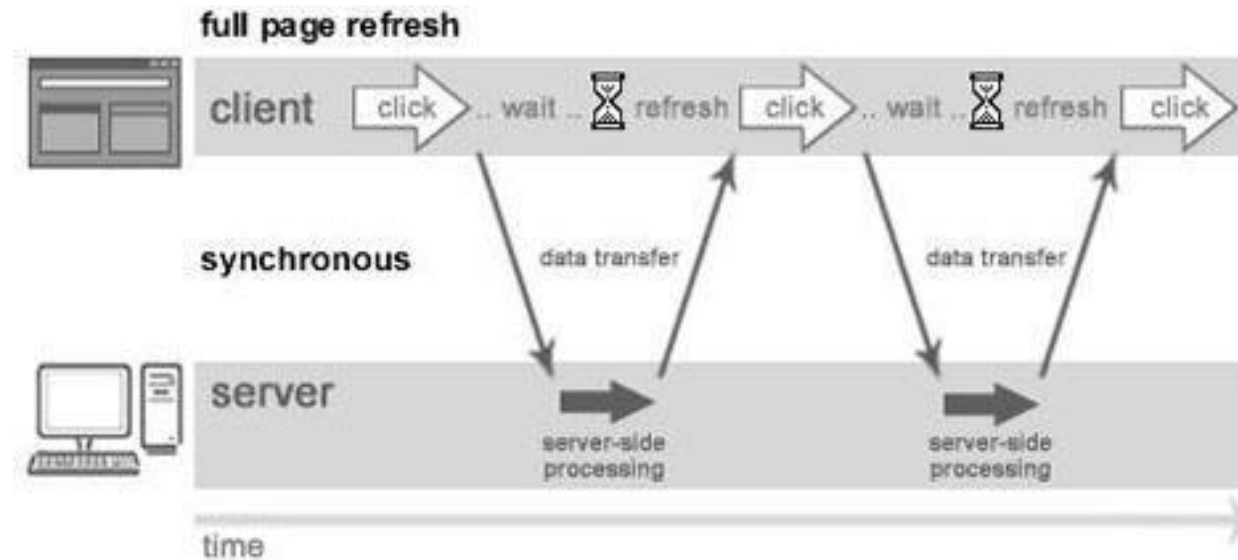
Ajax

- ◆ Overview Ajax
- ◆ Implementation Ajax in jQuery
- ◆ jQuery Ajax
 - ◆ \$.ajax(options)
 - ◆ \$.post(), \$.get()
 - ◆ \$.getJSON()
 - ◆ \$.getScript()
 - ◆ \$(image).load()

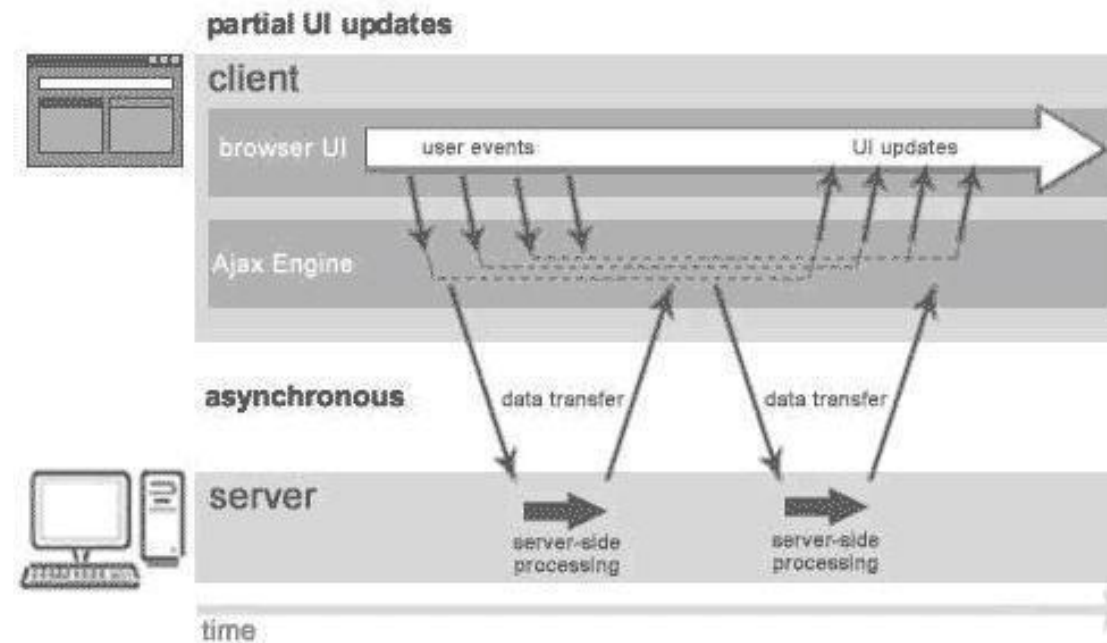
- ◆ AJAX = **A**synchronous **J**avaScript **A**nd **X**ML
 - ◆ Asynchronous loading
 - ◆ Data representation in XML
 - ◆ Client side script: JavaScript
- ◆ Different representation of data
 - ◆ HTML
 - ◆ Text
 - ◆ JSON
 - ◆ JavaScript

Asynchronous loading

◆ Synchronous



◆ Asynchronous



- ◆ Ajax
 - ◆ `$.ajax(options)`
- ◆ Ajax POST
 - ◆ `$.post()`
- ◆ Ajax GET
 - ◆ `$.get()`
- ◆ Ajax POST | GET with JSON
 - ◆ `$.getJSON()`
- ◆ Ajax with SCRIPT
 - ◆ `$.getScript()`

\$.ajax(options)

Example

Syntax

```
$.ajax({  
    url: "", //--địa chỉ server  
    data: {}, //--tham số truyền đến server  
    success: function (response) {...}, //-- xử lý phản hồi từ server  
    type: "", //--phương thức truyền: GET, POST, PUT, DELETE...  
    dataType:"" //--kiểu của dữ liệu nhận từ server text, xml, json...  
})
```

```
$.ajax({  
    url: "/Ajax/Search",  
    data: {min:5, max:10},  
    success: function (response) {  
        alert(response);  
    }  
})
```

Return data in Text

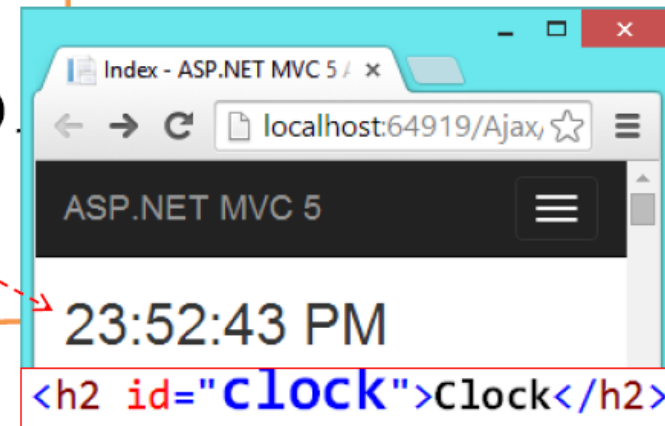
```
public ActionResult ServerTime()  
{  
    var text = DateTime.Now.ToString("HH:mm:ss tt");  
    return Content(text);  
}
```

1

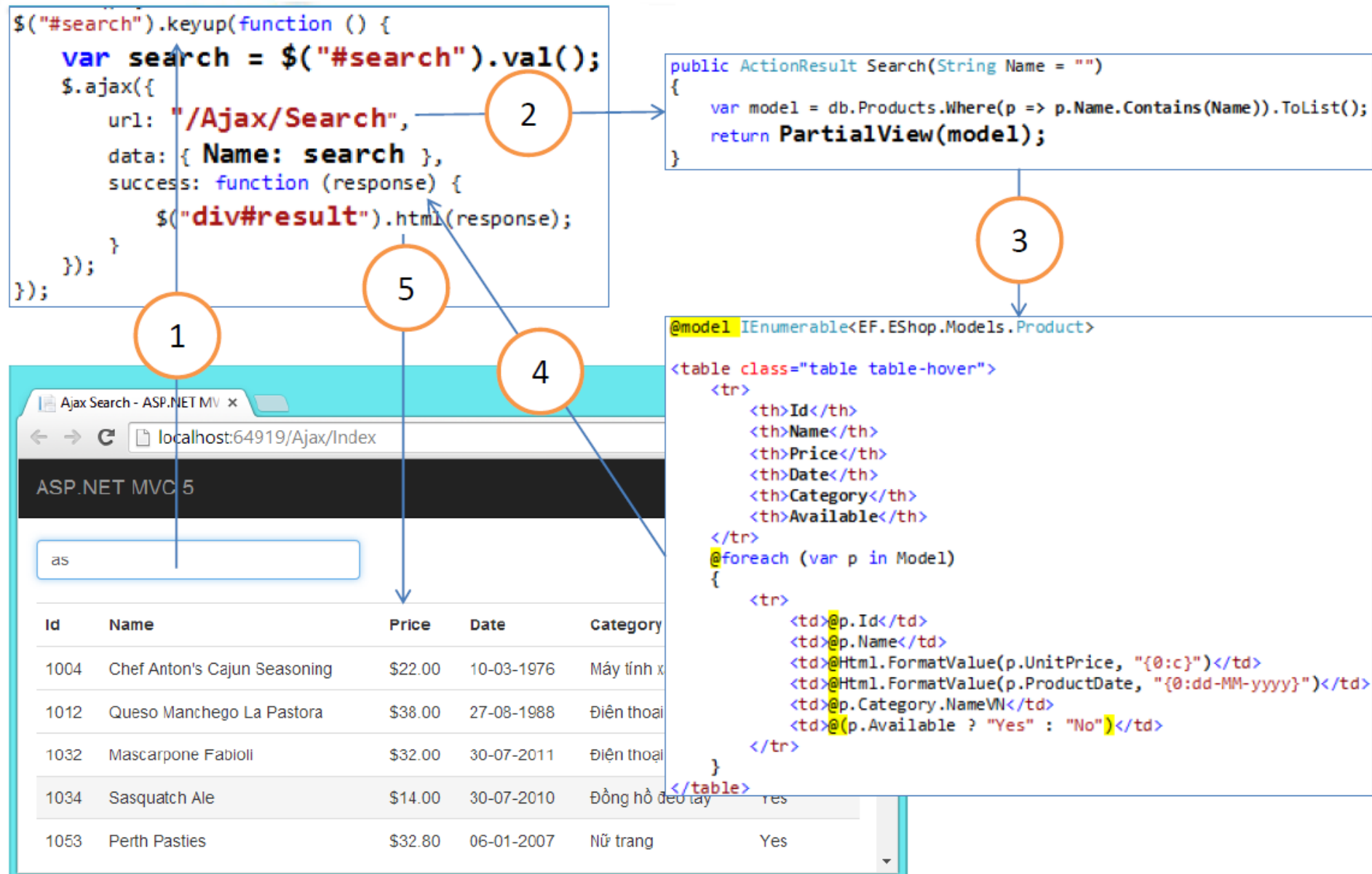
2

```
$(function () {  
    setInterval(function () {  
        $.ajax({  
            url: "/Ajax/ServerTime",  
            success: function (response) {  
                $("h2#clock").html(response);  
            }  
        }, 1000);  
    });  
});
```

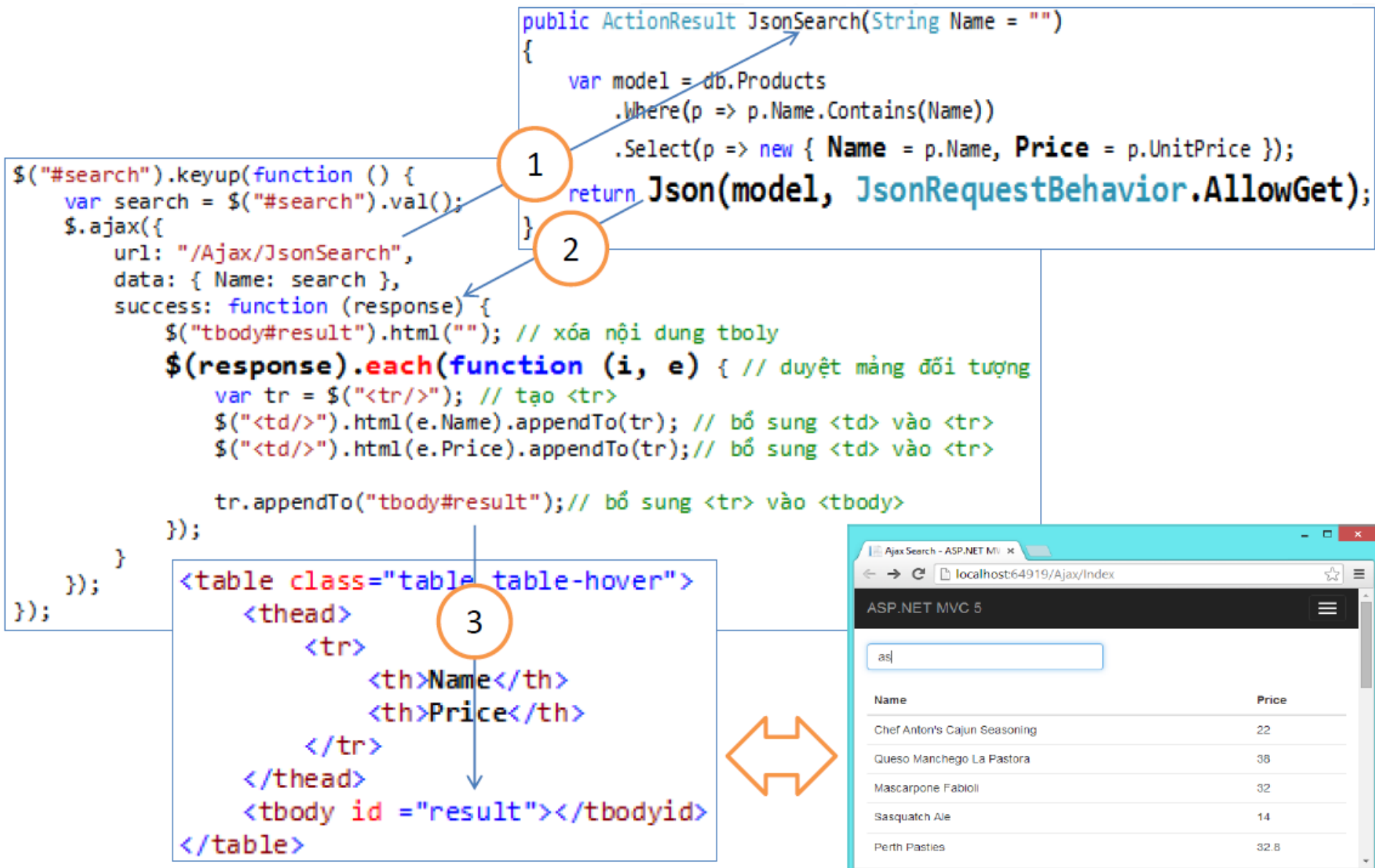
3



Receive data in Html



Receive data in JSON



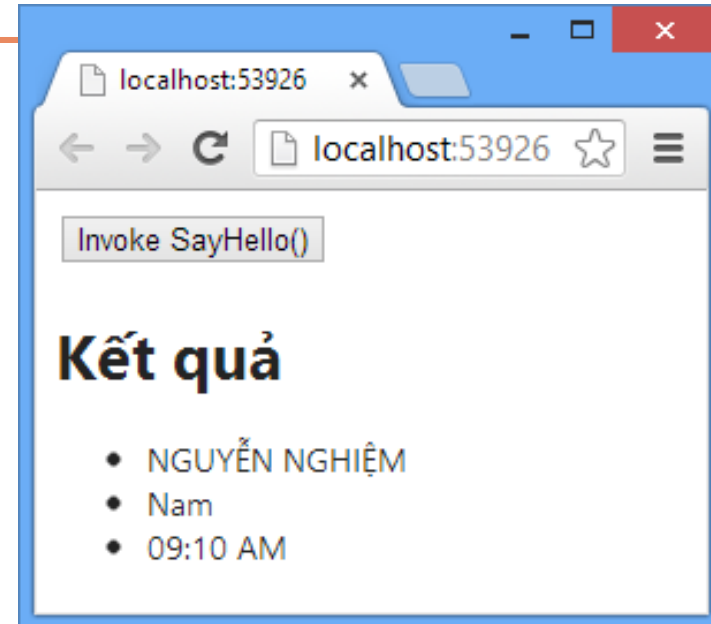
Some different cases of \$.ajax

- ◆ \$.post(url, data, function (response) { }, dataType)
 - ◆ \$.ajax({...type:"POST"...})
- ◆ \$.get(url, data, function (response) { }, dataType)
 - ◆ \$.ajax({...type : "GET"...})
- ◆ \$.getJSON(url, data, function (response) { })
 - ◆ \$.ajax({...type : "GET", dataType:"JSON"...})
- ◆ \$.getScript(url, data, function (response) { })
 - ◆ \$.ajax({...type : "GET", dataType:"JavaScript"...})

\$.getJSON

```
@section Scripts{
<script>
$(function () {
    $(".button").click(function () {
        $.getJSON(
            "Ajax/SayHello",
            "name=Nguyễn Nghiệm&gender=true",
            function (response) {
                $("#name").html(response.HoTen);
                $("#gender").html(response.GioiTinh);
                $("#time").html(response.ThoiDiem);
            }
        );
    });
});
</script>
}
<input type="button" value="Invoke SayHello()" />
<h1>Kết quả</h1>
<ul>
    <li id="name"></li>
    <li id="gender"></li>
    <li id="time"></li>
</ul>
```

```
public ActionResult SayHello(String name, bool gender)
{
    var info = new
    {
        HoTen = name.ToUpper(),
        GioiTinh = gender ? "Nam" : "Nữ",
        ThoiDiem = DateTime.Now.ToString("hh:ss tt")
    };
    return Json(info, JsonRequestBehavior.AllowGet);
}
```



Load image

◆ `$("selector").load(function() {})`

```
<button id="loadImage">Load Image</button>
```

```
<div id="images"></div>
```

```
@section Scripts{
```

```
<script>
```

```
    var i = 0;
```

```
    $(function () {
```

```
        $("#loadImage").click(function () {
```

```
            $("<img width=50/>").attr("src", "/Images/" + i + ".png")
```

```
            .load(function () {
```

```
                $(this).appendTo("#images").hide().show(1000);
```

```
                i = (i + 1) % 6;
```

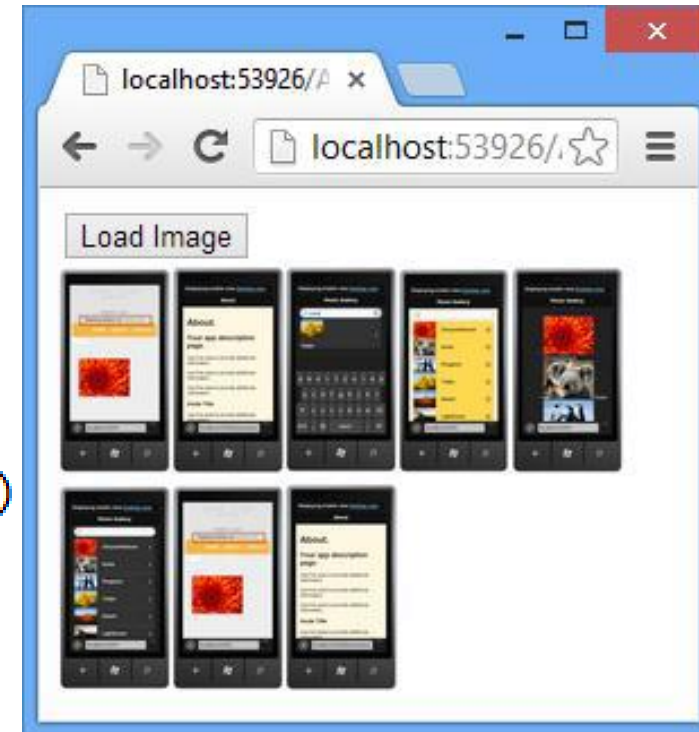
```
            });
```

```
        });
```

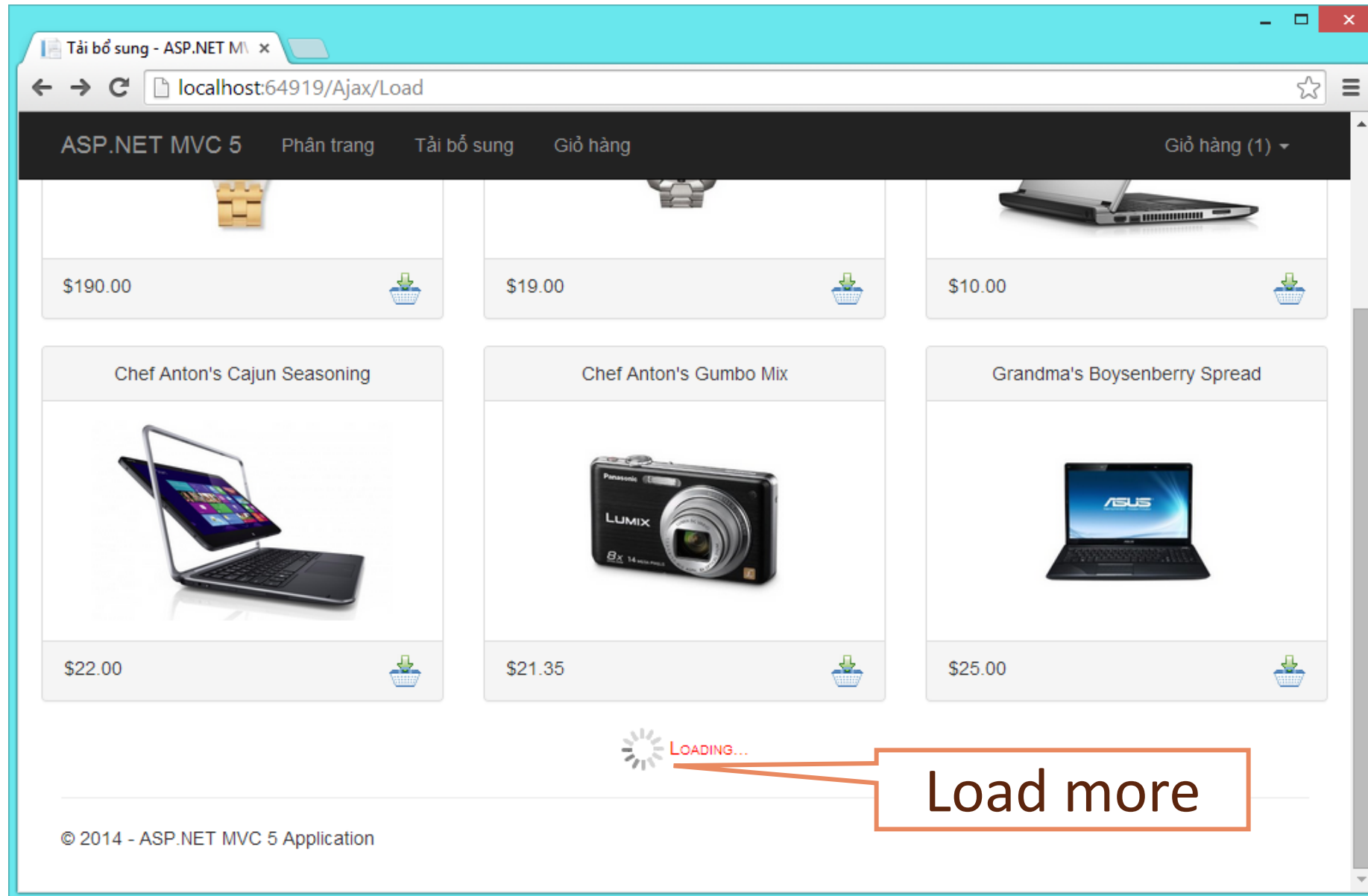
```
    });
```

```
</script>
```

```
}
```



Load more



Load more

```
$(function() {  
    // Tải trang đầu tiên  
    fnLoadMore();  
  
    // Tải thêm khi thanh cuộn chạm đáy trang  
    $(window).scroll(function() {  
        // Nếu cuộn đã chạm đáy  
        if($(window).scrollTop() + $(window).height() == $(document).height())  
        {  
            // Nếu còn hàng  
            if (pageNo < pageCount - 1) {  
                fnLoadMore();  
            }  
        }  
    });  
});
```

```
<div class="row" id="results"></div>  
  
<div class="loading">  
     Loading...  
</div>
```

```
var pageNo = 0;  
var pageCount = @(ViewBag.pageCount);  
  
function fnLoadMore(){  
    if($('.loading').css("display") == "none"){  
  
        $('.loading').show(); // hiển thị ảnh trạng thái  
  
        $.ajax({  
            url: "/Ajax/Load",  
            data: { pageNo: pageNo },  
            success: function (response) {  
                // bổ sung hàng tải được vào #results  
                $("#results").append(response);  
  
                pageNo++; // tăng trang cần tải tiếp theo lên 1  
                $('.loading').hide(); // ẩn ảnh trạng thái  
            }  
        });  
    }  
}
```

Load more

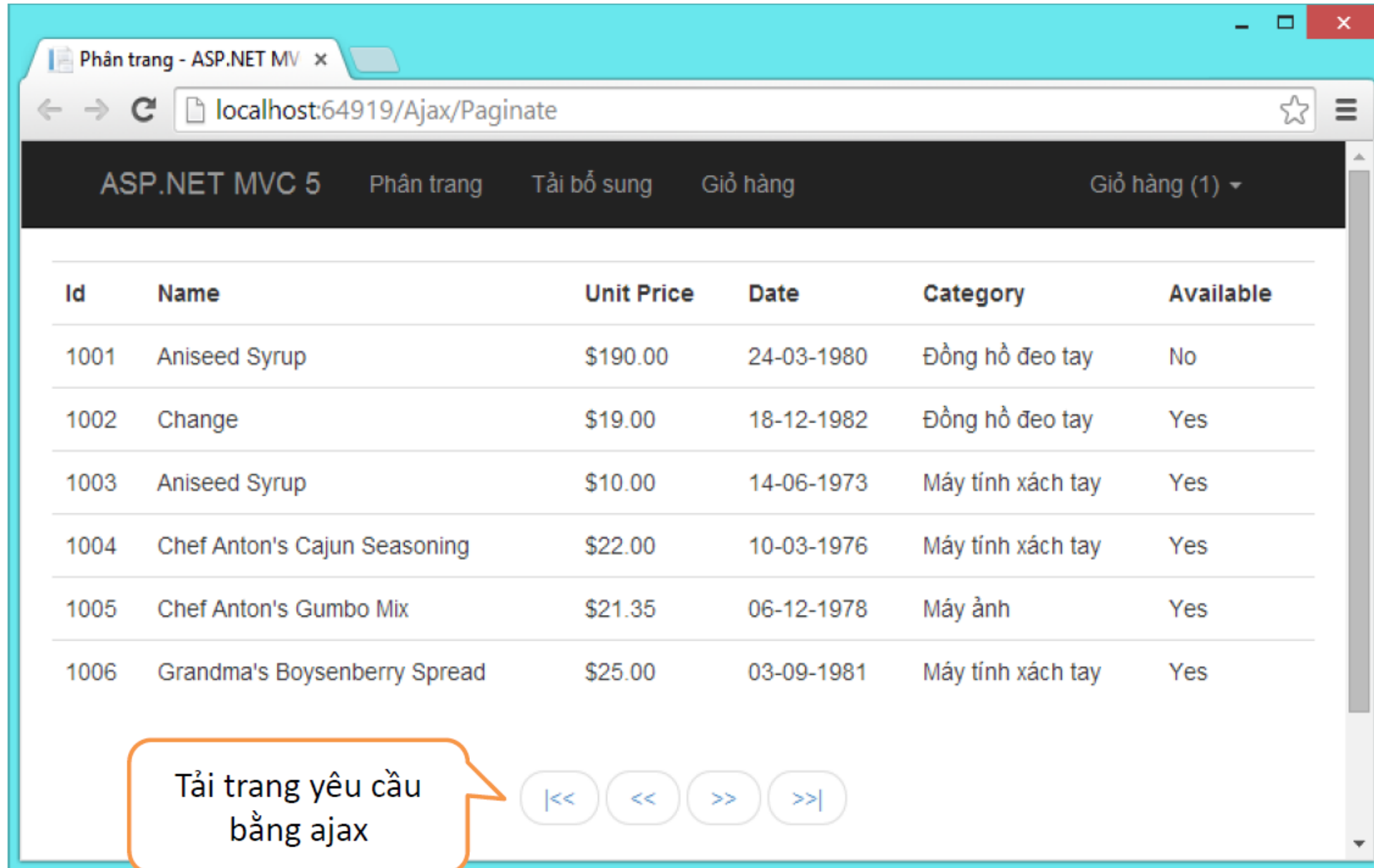
```
EShopV10 db = new EShopV10();

public ActionResult Index()
{
    var list = db.Products.ToList();
    ViewBag.pageCount = Math.Ceiling(1.0 * list.Count / pageSize);
    return View();
}

public ActionResult Load(int pageNo = 0, int pageSize = 6)
{
    var list = db.Products.ToList();
    var model = list.Skip(pageNo * pageSize).Take(pageSize);
    return PartialView(model)
}
```

```
@model IEnumerable<EF.EShop.Models.Product>
@foreach (var p in Model)
{
    <div class="col-md-4">
        <div class="panel panel-default nn-prod">
            <div class="panel-heading">@p.Name</div>
            <div class="panel-body">
                
            </div>
            <div class="panel-footer">
                <div class="row">
                    <div class="col-md-3">@p.UnitPrice</div>
                    <div class="col-md-9" style="text-align:right">
                        
                    </div>
                </div>
            </div>
        </div>
    </div>
}
</div>
```


Pagination with Ajax



The screenshot shows a web browser window with the title "Phân trang - ASP.NET MV" and the address bar displaying "localhost:64919/Ajax/Paginate". The page features a dark navigation bar with links for "ASP.NET MVC 5", "Phân trang", "Tải bổ sung", and "Giỏ hàng", along with a shopping cart icon showing "Giỏ hàng (1)". Below the navigation bar is a table with six columns: "Id", "Name", "Unit Price", "Date", "Category", and "Available". The table contains six rows of product data. At the bottom of the page, there are four pagination buttons: "<<|", "<<", ">>", and ">>|". An orange callout box with a pointer to the first two buttons contains the text "Tải trang yêu cầu bằng ajax".

Id	Name	Unit Price	Date	Category	Available
1001	Aniseed Syrup	\$190.00	24-03-1980	Đồng hồ đeo tay	No
1002	Change	\$19.00	18-12-1982	Đồng hồ đeo tay	Yes
1003	Aniseed Syrup	\$10.00	14-06-1973	Máy tính xách tay	Yes
1004	Chef Anton's Cajun Seasoning	\$22.00	10-03-1976	Máy tính xách tay	Yes
1005	Chef Anton's Gumbo Mix	\$21.35	06-12-1978	Máy ảnh	Yes
1006	Grandma's Boysenberry Spread	\$25.00	03-09-1981	Máy tính xách tay	Yes

Tải trang yêu cầu bằng ajax

Client code

|<< << >> >>|

```
<div class="row" id="results"></div>

<div class="loading">
   Loading...
</div>

<ul class="pager">
  <li><a href="0">|<<</a></li>
  <li><a href="--pageNo"><<</a></li>
  <li><a href="++pageNo">>></a></li>
  <li><a href="-1">>>|</a></li>
</ul>
```

```
<script>
  $(function() {
    // Tải trang đầu tiên
    fnLoadPage();

    // Tải trang tương ứng với nút được nhấp
    $(".pager a").click(function() {
      // tính lại trang cần tải
      eval("pageNo = (pageCount + " + $(this).attr("href") + ") % pageCount");
      fnLoadPage();
      return false;
    });
  });
</script>
```

```
var pageNo = 0;
var pageCount = @(ViewBag.pageCount);

function fnLoadPage(){
  // Hiển thị ảnh trạng thái
  $('.loading').show();

  // Tải trang bằng ajax
  $.ajax({
    url: "/Ajax/Paginate",
    data: { pageNo: pageNo },
    success: function (response) {
      // Hiển thị kết quả
      $("#results").html(response);

      // Ẩn ảnh trạng thái
      $('.loading').hide();
    }
  });
}
```

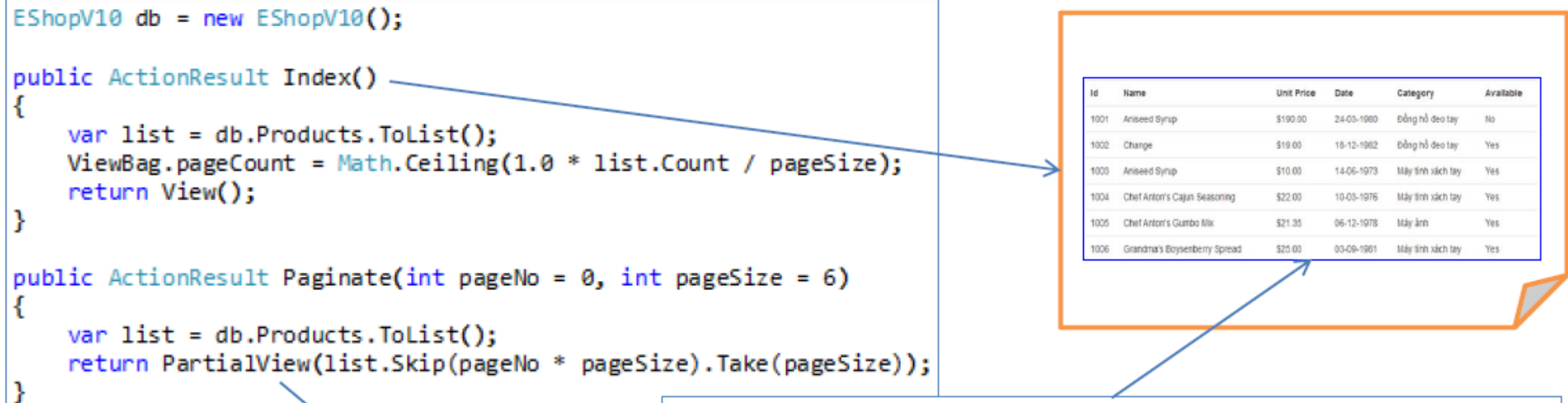
```
pageNo = (pageCount + 0) % pageCount;
pageNo = (pageCount + --pageNo) % pageCount;
pageNo = (pageCount + ++pageNo) % pageCount;
pageNo = (pageCount + -1) % pageCount;
```

Server code

```
EShopV10 db = new EShopV10();

public ActionResult Index()
{
    var list = db.Products.ToList();
    ViewBag.pageCount = Math.Ceiling(1.0 * list.Count / pageSize);
    return View();
}

public ActionResult Paginate(int pageNo = 0, int pageSize = 6)
{
    var list = db.Products.ToList();
    return PartialView(list.Skip(pageNo * pageSize).Take(pageSize));
}
```

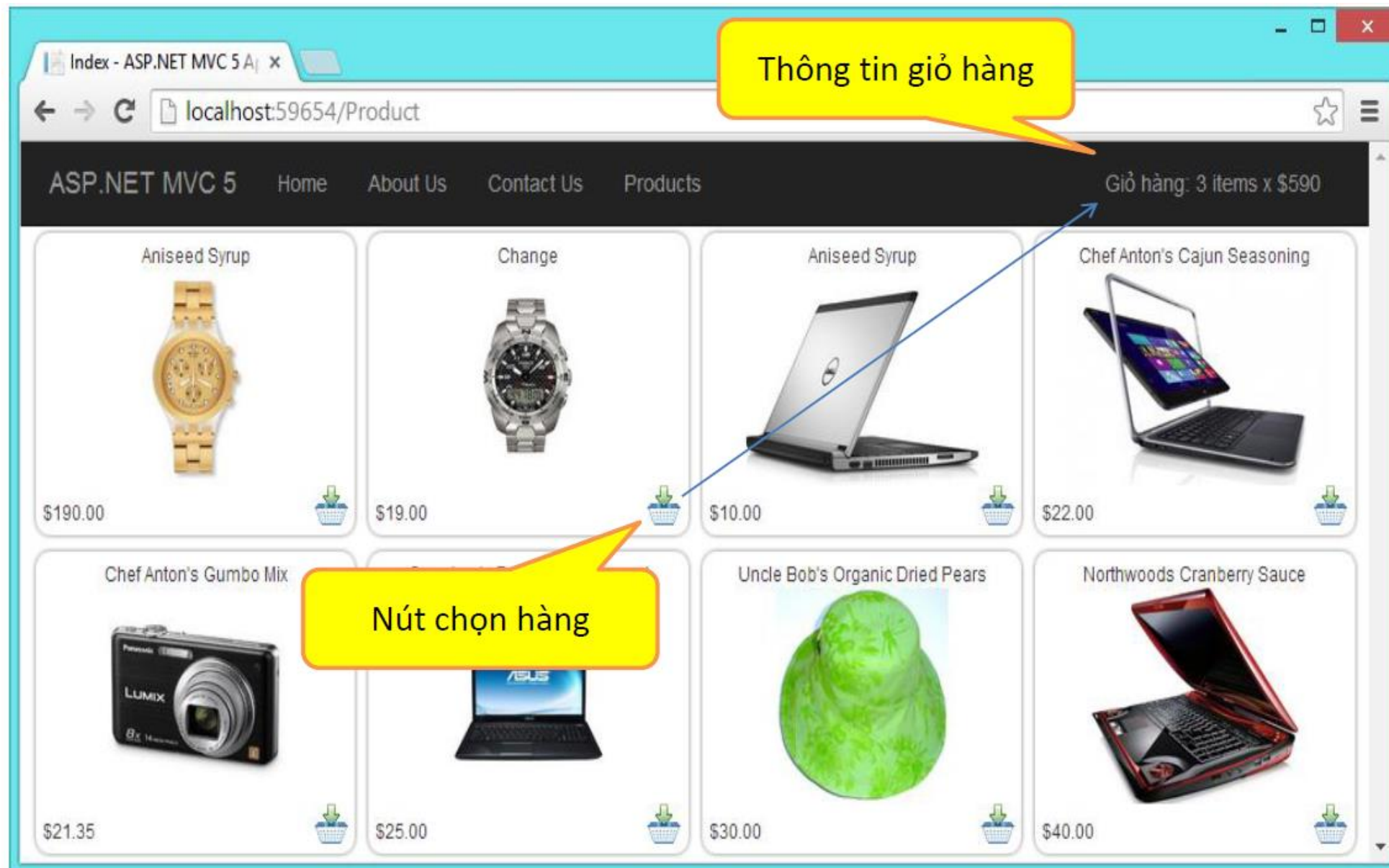


Id	Name	Unit Price	Date	Category	Available
1001	Aniseed Syrup	\$190.00	24-03-1980	Đồng hồ đeo tay	No
1002	Chang	\$19.00	18-12-1982	Đồng hồ đeo tay	Yes
1003	Aniseed Syrup	\$10.00	14-06-1973	Máy tính xách tay	Yes
1004	Chef Anton's Cajun Seasoning	\$22.00	10-03-1976	Máy tính xách tay	Yes
1005	Chef Anton's Gumbo Mix	\$21.35	06-12-1978	Máy ảnh	Yes
1006	Grandma's Boysenberry Spread	\$25.00	03-09-1981	Máy tính xách tay	Yes

```
@model IEnumerable<EF.EShop.Models.Product>

<table class="table table-responsive">
    <tr>
        <th>Id</th>
        <th>Name</th>
        <th>Unit Price</th>
        <th>Date</th>
        <th>Category</th>
        <th>Available</th>
    </tr>
    @foreach (var p in Model)
    {
        <tr>
            <td>@p.Id</td>
            <td>@p.Name</td>
            <td>@Html.FormatValue(p.UnitPrice, "{0:c}")</td>
            <td>@Html.FormatValue(p.ProductDate, "{0:dd-MM-yyyy}")</td>
            <td>@p.Category.NameVN</td>
            <td>@(p.Available?"Yes":"No")</td>
        </tr>
    }
</table>
```

Shopping card



Shopping card

The screenshot shows a web browser window titled "Shopping Cart Management" with the address bar displaying "localhost:59654/Cart/Index". The application has a dark navigation bar with links for "ASP.NET MVC 5", "Home", "About Us", "Contact Us", and "Products". On the right of the navigation bar, it says "Giỏ hàng: 3 items x \$590".

The main content area is titled "Shopping Cart Management" and contains a table with the following columns: "Name", "Unit Price", "Quantity", and "Total Price". The table lists three items:

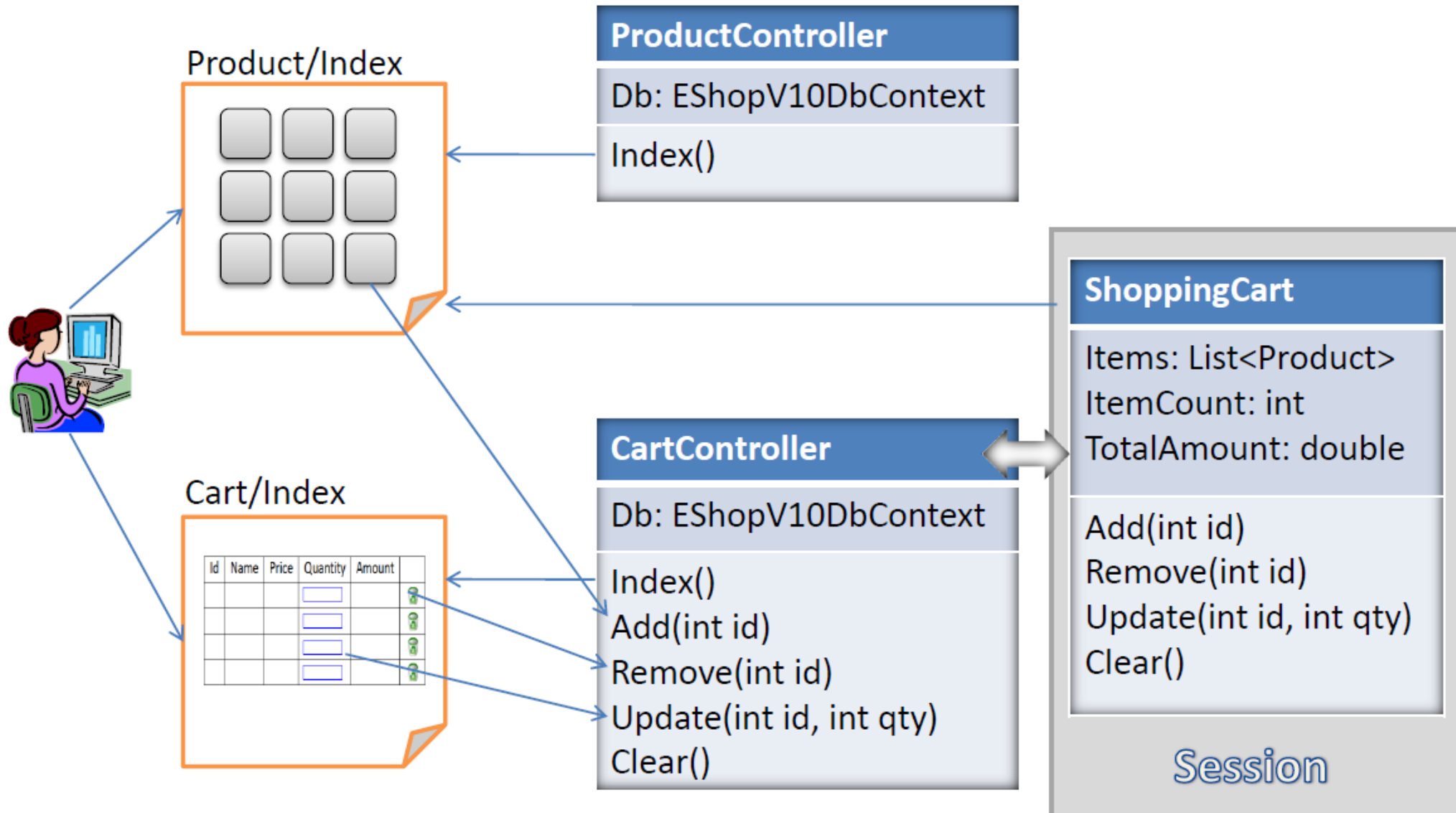
Name	Unit Price	Quantity	Total Price
Aniseed Syrup	\$190.00	2	\$380.00
Change	\$19.00	10	\$190.00
Aniseed Syrup	\$10.00	2	\$20.00

Below the table are four buttons: "Update", "Clear", "Continue", and "Checkout". At the bottom left, it says "© 2014 - ASP.NET MVC 5 Application".

Two yellow callout boxes with orange borders provide annotations in Vietnamese:

- A box labeled "Xóa" (Delete) with an arrow pointing to the trash icon of the first item.
- A box labeled "Cập nhật số lượng" (Update quantity) with an arrow pointing to the quantity input field of the second item.

Shopping card



Order

Checkout - ASP.NET MVC x

localhost:59654/Order/Checkout

ASP.NET MVC 5 Home About Us Contact Us Products Giỏ hàng: 3 items x \$590

Thông tin đơn đặt hàng

Khách hàng	nnghiem2014
Ngày đặt	6/17/2014 10:33:00 PM
Địa chỉ nhận	105 Bà Huyện Thanh Quan
Tổng tiền	590
Ghi chú	Cần chuyển hàng đúng thời hạn

Purchase

Chi tiết đơn đặt hàng

Tên hàng hóa	Đơn giá	Số lượng	Thành tiền
Aniseed Syrup	\$190.00	2	\$380.00
Change	\$19.00	10	\$190.00

Orders

OrderDetails